NATURAL ENGLAND

LIFE WADER

WETLAND CASE STUDY

Coastal wetland interventions in Northumberland to reduce diffuse pollution

Caitlin Dolan, Liz Humphreys, Mark Renfrey

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WETLAND CASE STUDY

THE PROBLEM

Many of the water courses in the Tweed catchment are loaded with high levels of phosphates and nitrates. These high levels originate from a range of human activities including agriculture, sewage systems, transport, mining and industry. An excess in nutrients can cause a host of environmental issues that harm the ecosystem such as: eutrophication, dead zones/oxygen depletion, human health hazards, and biodiversity loss.



Budle Bay is a Site of Special Scientific Interest (SSSI) in Northumberland where algal blooms have become present in the bay due to elevated nutrient levels. It is an important site for a number of bird species that use the bay to breed, feed, and rest. As farmland surrounds the bay, there is a direct pathway for nutrients to flow from fields into the bay. Therefore, this this coastal stretch was chosen as a WADER priority site for interventions.

THE RESPONSE

WADER is working with local farmers and land managers to install wetland systems around Budle Bay to help stem nutrient and sediment outflow into the watercourse. Wetlands and Sustainable Drainage Systems are simple and effective ways to improve water quality and biodiversity by tackling diffuse pollution. Wetlands have numerous environmental benefits as they filter and capture nutrients, slow water flow, store carbon, and create important habitats for wildlife- boosting biodiversity.

LOCATE

Environment Agency data was used to create Nitrogen and Phosphate hotspot maps to highlight priority locations.

OPTIONS

Locations were ground truthed for suitable sites, land owners engaged and options for possible interventions listed.

DESIGN

Qualified design team was hired to undertake any required surveys and to create planning ready designs.

PERMISSIONS

Engaged with local council to seek planning permission where required.

BUILD

Construction team was hired to work alongside designers and project team to complete the build.

LOCATION

The Waren Mill site is located within the Lindisfarne SSSI and encapsulates farmland leading directly to the saltmarsh habitat at Budle Bay. There is a small corridor of bracken bankside land between the fields and saltmarsh that is rented to and managed by the National Nature Reserve. In this area a total of six field drains were located as sites for wetland interventions. Five of these drains flow directly into the bay and the most northernly drain is located at an existing ditch and small reedbed.

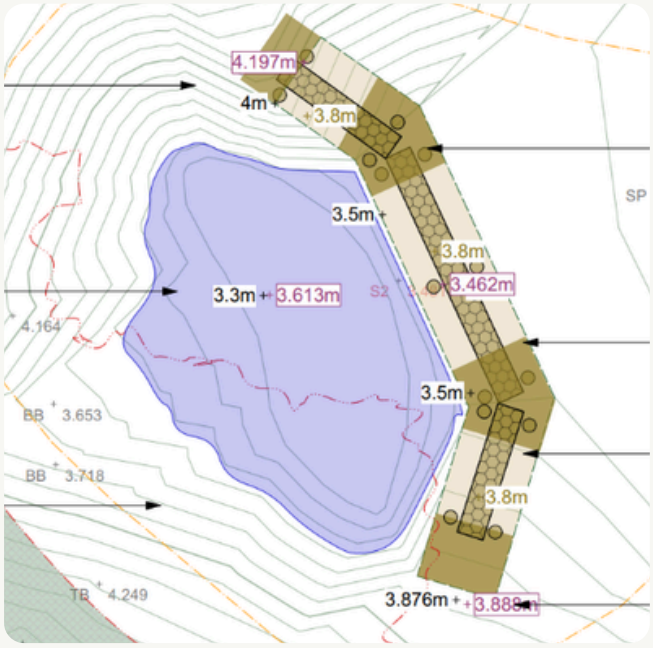


PLANNING PERMISSION

The work at this site did not require planning permission as it is part of the previously agreed SSSI management plan and has undergone habitat assessments.

DESIGN PROCESS

Specialist wetland designers Blue Green Studios were hired to design interventions that would work to reduce diffuse pollution whilst protecting the sensitive nature of the site. It was decided that multiple smaller scale interventions would be more effective and appropriate at this site. Simple bunded settlement ponds with coir roll seepage barriers were designed for the field drains that drained directly to the bay. At the previous reedbed site, enhancements were designed to create a multi-cell pond system flowing through coir roll seepage barriers and a leaky damn.



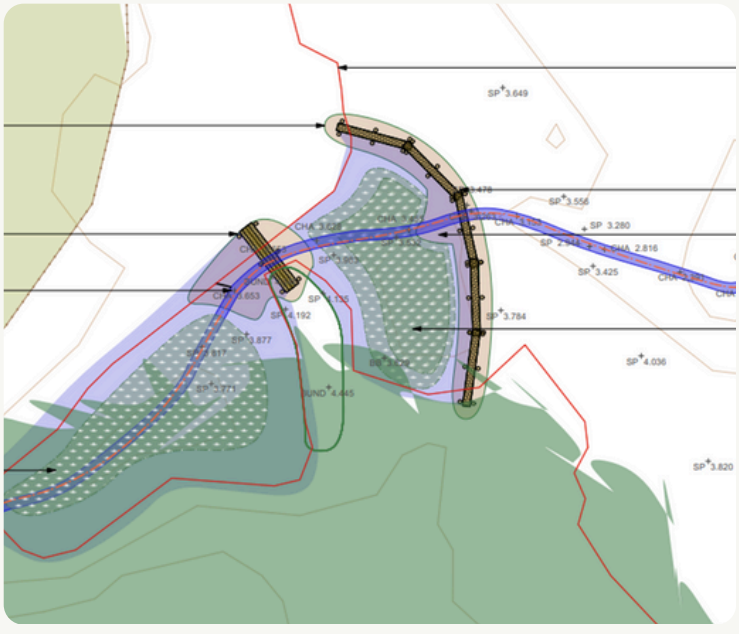


Figure 1. Design examples from Green Blue Studios for the Waren Mill Wetland Site.

CONSTRUCTION

A local construction company with wetland building experience and knowledge of the local topography, Haywood Contracting, were employed to build the designed interventions. The designs were used as a guide, with necessary adaptations made during construction where the land required design tweaks to gain the best outcomes. Environmentally sensitive construction methods were used and strict timings were adhered to in order to minimise disturbance to vulnerable bird species.

In 2024 interventions were placed at the reedbed site (3 pond system built) and at 3 field drains (1 double pond system and 2 single ponds built). After being trialled for a year, in 2025 these interventions were expanded in size and 2 more field drains were located (3 ponds built). In total across the site **5 wetland/ settlement pond systems** containing a total of **10 ponds were constructed**.



Figure 2. Construction of a settlement pond in process.



Figure 3. Typical area of bracken bankside available for construction.



Figure 4. Typical example of settlement pond with staked coir roll seepage barrier.

INTERVENTION RESULTS

REEDBED SITE

Grid Reference: NU14093 35437
Constructed 09.2024, Enhanced 08.2025

At the reedbed site a leaky dam was placed at the top of the ditch. This then flows into a 3 pond system with coir roll seepage barriers and a bunded bottom pond. In 2025 the top and bottom ponds were expanded to tackle overtopping during storm events. Visual algal growth in the top pond that is clear by the bottom pond has shown a good indication of the system effectively filtering nutrients. Reeds have began to naturally re-establish across the ponds.

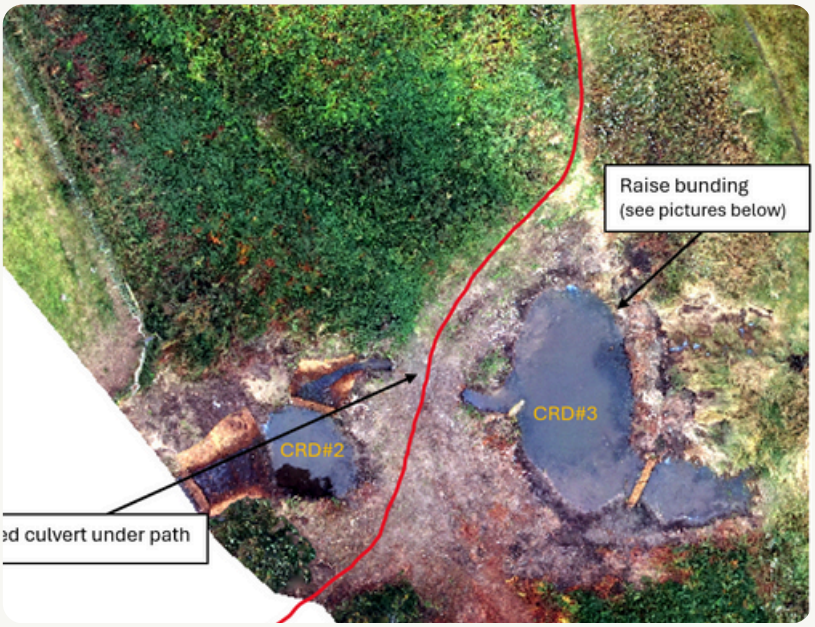


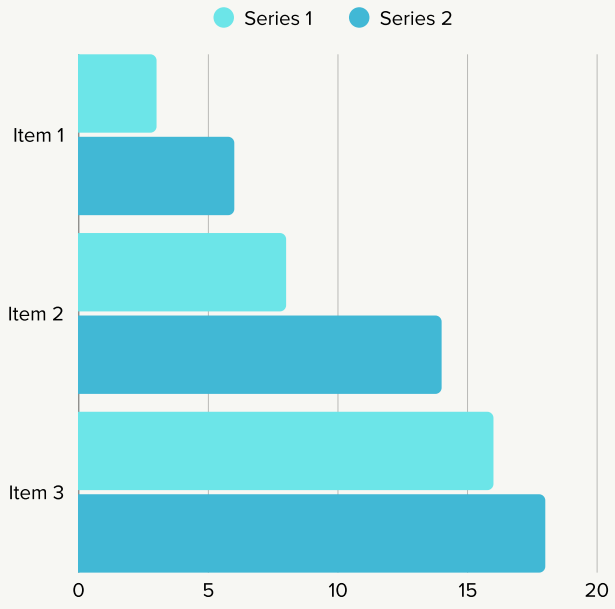


Figure 5. Reducing algal blooms from pond 1 to 3 at the reedbed site 12.06.2025

MONITORING

Monthly monitoring was conducted for *years at this site with lab tests conducted for Nitrogen and Phosphate Levels. Results show *xyz . This indicates that the system is successfully reducing the nutrient load across the ponds, resulting in less nutrients entering the bay.

Sediment testing has also shown increased nutrient levels in the ponds that may be removed during maintenance and reapplied to farmland.

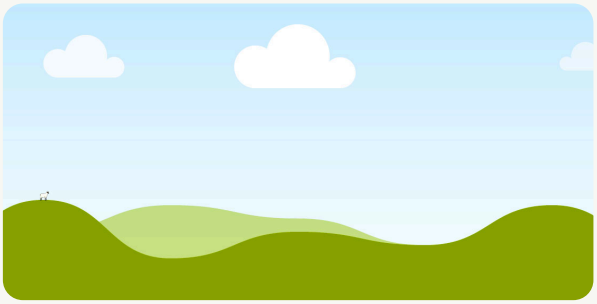


Settlement Pond 1

Grid Reference: NU 14267 35235
Constructed 09.2024, Enhanced 08.2025

Settlement pond 1 consists of a 2 pond system that flows through a coir roll. The top pond intercepts clay field drains, and a larger lower pond discharges away from the previous incised channel.

In 2025 both ponds were expanded to increase retention capacity.





Settlement Pond 2

Grid Reference: NU1433135122
Constructed 09.2025

Settlement pond 2 consists of a 3 pond system that flows through a coir roll. The ponds either side capture a field drain each and there is an overflow pond in the middle to increase retention when high flow occurs.





Settlement Pond 3 & 4

Grid Reference: NU 14503 34842, NU 14547 34783

Constructed 09.2024, Enhanced 08.2025

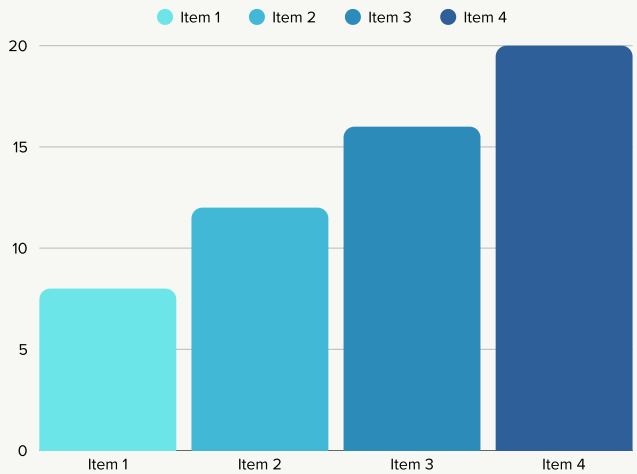
Settlement ponds 3 and 4 had less space for interventions and more unstable soils. Therefore a single pond intervention was placed at each to capturing runoff from field drains.

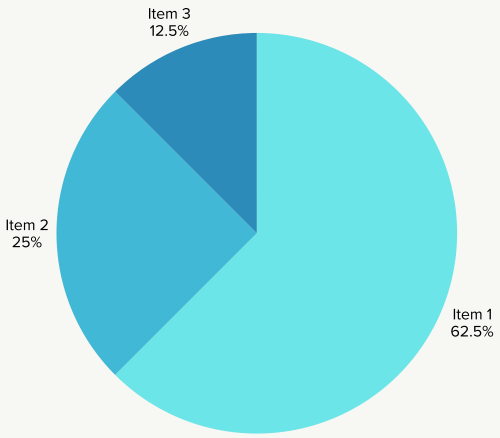




MONITORING

Unfortunately we did not have capacity or funding to conduct lab analysis across all of the ponds at Waren Mill, therefore the reedbed site was chosen as the best intervention to monitor to a high level of accuracy with lab testing. Monthly spot testing of water levels and nutrients (Nitrogen and Phosphate) was conducted with handheld Hanna probes for one year across all of the remaining settlement ponds. These tests were quick, and although less accurate at lower concentrations than lab tests, they were suitable to use for maintenance purposes to indicate if any critical errors may have arose in the system. If that occurred, high accuracy lab tests could be done to confirm findings and appropriate adaptations to the system made.





CASE STUDY 2 - KILN POINT

LOCATION

The Kiln Point site is located on the eastern side of Budle Bay just outside of Lindisfarne SSSI. The site captures a farmland ditch which leads directly into Budle Bay. There is a clear algal trail at the ditch outlet. The site captures runoff from the surrounding farmland including a precipitation fed ditch and several field drains and old well connections.



PLANNING PERMISSION

After liaising with Northumberland County Council it was determined that planning permission was not necessary as the below conditions were met:

1) On a commercial agricultural holding of **more than 5ha**, and not sited on a separate parcel of l and of less than 1ha.
2) Created for the **management of the land** for agricultural purposes (including nutrient capture/ related environmental projects).
3) Does not **exceed 0.1ha** in area.

DESIGN PROCESS

Specialist wetland designers Blue Green Studios were hired to design interventions that would work to reduce diffuse pollution at the site. A multiple pond system incorporating swales and coir rolls was designed to slow the passage of water and increase the potential for the uptake of nutrients and the settlement of sediment to reduce pollution into Budle Bay. It was important that the design was both simple and effective to ensure maintenance for this site was manageable for the farmer.

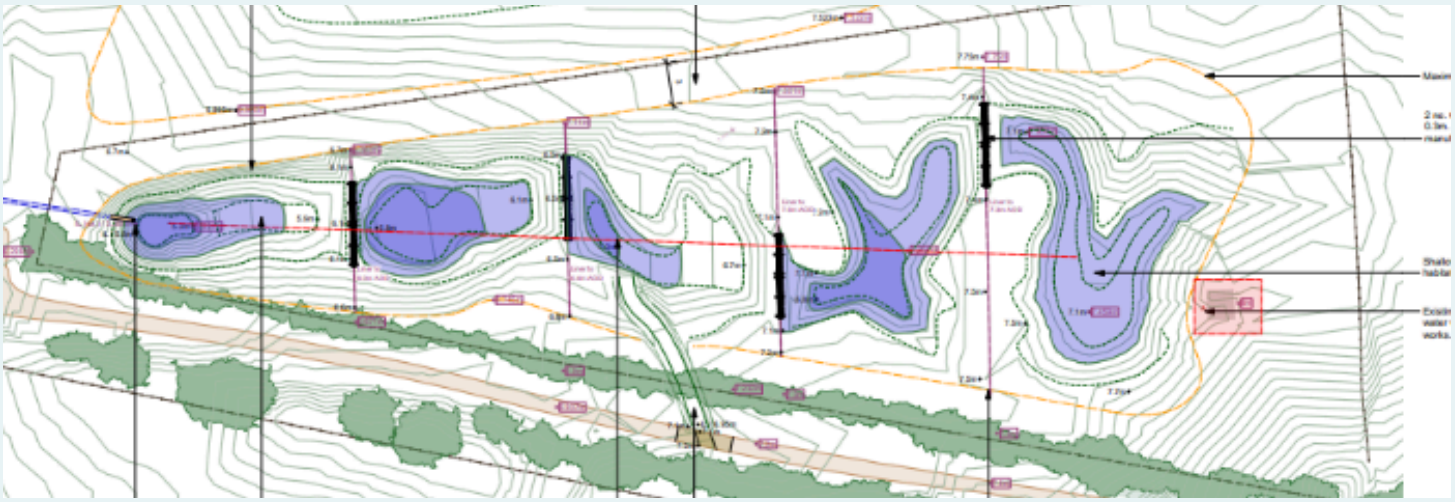


Figure 6. Design examples from Green Blue Studios for the Kiln Point Wetland Site.

CONSTRUCTION

A local construction company with wetland building experience and knowledge of the local topography, Haywood Contracting, were employed to build the designed interventions. The designs were used as a guide, with necessary adaptations made during construction where the land required design tweaks to gain the best outcomes. Environmentally sensitive construction methods were used and strict timings were adhered to in order to minimise disturbance to vulnerable bird species.

In 2025 a **5 pond wetland system** was constructed at the site.



Figure 7. Construction area for Kiln Point Wetland



Figure 8. Example of the excavator creating curved edges to create gradual gradient at the edges of the ponds.



Figure 9. Example of the stone swale installation.

KILN POINT WETLAND

Grid Reference: NU15409 35269
Constructed 09.2025

Five shallow, asymmetric settlement ponds were created with coir rolls in between each one. Additional clay lining was not required as the sediment on site was clay-based and able to hold water sufficiently once compacted. Swales lined with stones were also installed as an additional method to slow water passing between the ponds and to trap sediment. The water holding capacity of the area was greatly increased, providing important freshwater habitat that will add conservation value to the area.





Figure 10. Freshwater habitat created

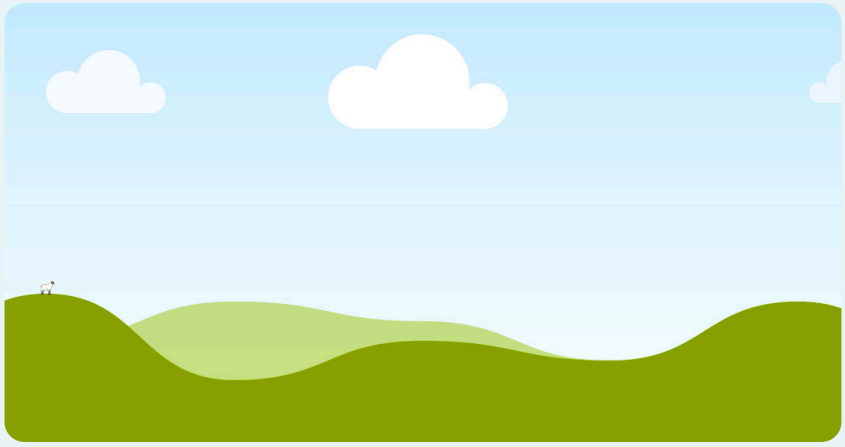
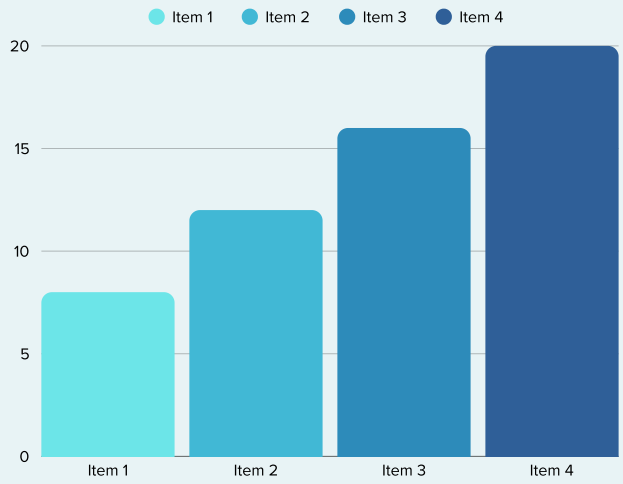


Figure 11.

MONITORING

Monthly monitoring was conducted for *years at this site with lab tests conducted for Nitrogen and Phosphate Levels. Results show *xyz . This indicates that the system is successfully reducing the nutrient load entering the bay.



Challenges and Lessons Learnt